

Troubleshooting Guide: Slug Catcher — Liquid Level Surge from Pipeline Slugging

Category: Multiphase Flow / Flow Assurance **Unit:** Finger-Type Slug Catcher, receiving multiphase pipeline flow **Tools:** Transient multiphase pipeline flow simulation (e.g., OLGA) compared against actual slug catcher level trend data **Fluid Package:** PR, used for the compositional multiphase pipeline model within the transient flow simulator **Symptom:** Intermittent liquid level surges in the slug catcher beyond normal range, with occasional high-level trips risking liquid carryover to downstream gas compression

1. Quick Reference

Item	Detail
Reported symptom	Slug catcher liquid level surging intermittently beyond normal operating range; occasional high-level trips
Initially unclear	Whether this was a level control/instrumentation issue or a genuine change in pipeline slugging behavior
Actual root cause	Increased slug volume/frequency caused by operating in a low gas-rate turndown regime that promoted severe slugging at a low point in the pipeline — a regime not accounted for in the current operating envelope
Fix	Adjusted gas throughput/pigging frequency; revised minimum flow operating guideline based on transient simulation results
Diagnostic signal	Transient multiphase simulation predicted significantly larger slug volumes at the current low-flow operating point than at the previous higher-flow regime, matching the observed level surge pattern
Prevention	Use transient simulation to define a safe operating envelope for turndown scenarios; periodic pigging schedule; revisit level control surge margin

2. Symptom

- **Slug catcher liquid level surging intermittently** well beyond its normal operating range.
- **Occasional high-level trips**, creating risk of **liquid carryover to downstream gas compression** — a significant reliability and potential equipment-damage concern.

3. Why This Wasn't Assumed to Be a Level Control Problem

Level surges and trips are often first attributed to level control tuning or instrument issues — a relatively simple fix if true. But slug catchers are specifically designed to absorb the **normal, expected** liquid slugs generated by multiphase pipeline flow; if the *slugs themselves* have grown larger or more frequent than the design/operating assumption, no amount of level control retuning will fully solve the problem, since the fundamental liquid inventory arriving in each slug has changed. It was necessary to determine whether the **pipeline's multiphase flow behavior itself** had shifted.

4. Diagnostic Approach

Step 1 — Review recent operating history for changes

Recent operating conditions were reviewed, and a notable change was identified: the pipeline had recently been operating at a **reduced gas throughput (turndown)** relative to its more typical rate.

Step 2 — Run a transient multiphase pipeline flow simulation

A transient multiphase flow model of the pipeline (e.g., in OLGA) was built/updated, using **PR** for compositional fluid properties, to predict expected slug volume and frequency across a range of gas flow rates — including the current reduced-throughput operating point.

Why a transient (not steady-state) simulation is needed: slugging is inherently a dynamic, time-varying multiphase flow phenomenon. A steady-state process simulator (as used in most of the other cases in this series) cannot predict slug volume or frequency — that requires a transient, mechanistic multiphase flow model built specifically for pipeline hydraulics.

Step 3 — Compare predicted slug behavior across flow regimes

The simulation showed that at the **current reduced gas flow rate**, predicted slug volumes were **significantly larger** than at the pipeline's more typical higher-flow operating point — consistent with a known multiphase flow phenomenon where certain **low-flow regimes promote severe slugging** at low points in the pipeline profile (as liquid accumulates and is periodically pushed through by the gas phase in large slugs, rather than flowing more continuously at higher gas rates).

Step 4 — Confirm against actual slug catcher level data

The simulation's predicted slug volumes at the reduced-flow regime were consistent with the magnitude and timing pattern of the actual level surges observed at the slug catcher, confirming the mechanism.

Quantitative Basis

- Slug catcher design level band: 20-70% (alarm at 85%, trip at 95%). Level surges observed spiking intermittently to **92-97%**.
- Normal gas throughput: 145 MMscfd. A downstream compressor limitation had forced turndown to **58 MMscfd (40% of normal) for 3 weeks**.
- OLGA transient simulation predicted maximum slug volume of **180 bbl at 145 MMscfd**, rising to **640 bbl at 58 MMscfd — a 3.6x increase**, consistent with severe slugging onset at the pipeline's low point.

- Observed level surge magnitude (92–97%, consuming roughly 540–620 bbl of usable freeboard) closely matched the **640 bbl simulated slug volume** at the reduced flow rate.

5. Root Cause

Operating the pipeline at a reduced gas throughput (turndown) pushed the multiphase flow regime into a severe-slugging condition at a low point in the pipeline, generating larger and more frequent liquid slugs than the slug catcher’s normal operating assumptions accounted for — causing the observed level surges and high-level trips. This flow regime had not been explicitly evaluated as part of the established operating envelope.

6. Corrective Action

1. **Adjusted gas throughput** (and reviewed pigging frequency) to move operation away from the severe-slugging regime identified by simulation.
2. **Revised the minimum flow operating guideline**, informed directly by the transient simulation’s predicted slug volume/frequency across flow rates, rather than an assumption carried over from the original design case.

7. Verification

- Minimum flow guideline raised to **95 MMscfd**, the threshold above which simulation showed slug volume staying below 250 bbl.
- Level surges reduced to a maximum of **74%**, well within the 85% alarm threshold, over the following **60 days**.
- No further high-level trips observed under the corrected operating envelope.

8. Prevention / Long-Term Fix

- Established use of **transient multiphase simulation to define a safe operating envelope for turndown scenarios**, rather than relying on the original design case alone across the full range of possible operating rates.
 - Maintained a **periodic pigging schedule** to manage liquid inventory in the pipeline proactively.
 - Revisited **level control surge margin** at the slug catcher to ensure adequate response capability even within the confirmed safe operating envelope.
-

9. General Troubleshooting Checklist (Reusable)

- ☐ Confirm whether slug catcher level surges/trips correlate with a recent change in gas throughput or other pipeline operating conditions
- ☐ Before assuming a level control/instrumentation cause, consider whether the underlying multiphase flow regime itself may have changed
- ☐ Build or update a **transient multiphase flow model** of the pipeline (not a steady-state process simulator) to predict slug volume/frequency across the relevant range of flow rates
- ☐ Compare predicted slug behavior at the current operating point against the design/typical operating point
- ☐ Cross-check simulation predictions against actual slug catcher level trend data (magnitude and timing) to confirm the mechanism

- ☐ If a low-flow severe-slugging regime is identified, revise the minimum flow operating guideline rather than only retuning level control
- ☐ Use simulation results to define a documented safe operating envelope for turndown scenarios going forward
- ☐ Maintain pigging schedule and review slug catcher level control surge margin as complementary, not substitute, measures

10. Key Takeaway

A slug catcher is designed for the pipeline's **expected** slugging behavior — but that behavior isn't constant across all flow rates. Certain low-flow turndown regimes can trigger much larger, more severe slugs than higher, more typical flow rates, and no amount of level control retuning fixes that if the fundamental slug volume has grown. When level surges correlate with a throughput change, use transient multiphase simulation — not a steady-state model — to check whether the current operating point has moved into a severe-slugging regime, and define your safe operating envelope from that, not just from the original design case.

Related Concepts / Tags

slug-catcher multiphase-flow pipeline-slugging OLGA transient-simulation severe-slugging turndown flow-assurance pigging level-control

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.